Report on "Matrix Theory Reloaded: A BPS Road to Holography"

(Blair, Lahnsteiner, Obers, Yan)

Summary of the paper

The paper revisits the decoupling limits that define matrix theory on Dp-branes (BFSS for $p = 0$). It stresses that these limits are *BPS* limits, in which the brane tension is balanced against the RR charge. Applied to the whole type II theory, the limit gives "matrix p-brane theory" (MpT), with a non-Lorentzian target space carrying a codimension-$(p+1)$ foliation. The authors then make three main claims, in answer to three questions posed in the introduction.

1. Applying the *same* MpT limit to a background Dp-brane geometry produces the near-horizon limit. This recovers $\mathrm{AdS}_5 \times S^5$ and the Itzhaki–Maldacena–Sonnenschein–Yankielowicz (IMSY) geometries. Applying it to a misaligned Dq-brane ($q \neq p$) gives non-Lorentzian bulk geometries.

2. Each BPS limit corresponds to a DLCQ of M-theory. They conjecture that holography is a $\mathrm{DLCQ}^n/\mathrm{DLCQ}^m$ correspondence with $m > n$. $\mathrm{AdS}_5/\mathrm{CFT}_4$ is the case $\mathrm{DLCQ}^0/\mathrm{DLCQ}^1$.

3. Undoing the limit is a p-brane generalisation of the $T\bar{T}$ deformation, so the near-horizon bulk geometry arises as an intrinsic deformation of the flat non-Lorentzian asymptotic geometry. This yields new $T\bar{T}$-like flow equations, for example $\partial_\lambda \mathcal{L} = \frac{1}{2\lambda^2}\{\mathrm{tr}(1-\lambda\mathcal{T}) - (p-1)[\det(1-\lambda\mathcal{T})]^{1/(p-1)} - 2\}$. This flows free scalars to the Dirac–Nambu–Goto action. For $p = 1, 2$ there are also flows from Yang–Mills to DBI.

1 Principal achievement

The main achievement is a *unifying and constructive framework*. It shows that the matrix-theory decoupling limit and the Maldacena near-horizon limit are one and the same BPS limit, applied at infinity or to the brane geometry. This turns a historically heuristic relationship into a geometric statement. The explicit rewriting of the near-horizon metric as $\omega \to (r/\ell)^2$ in the MpT limit makes it transparent. The framework also gives a systematic recipe for generating candidate holographic duals, including non-Lorentzian ones. It reproduces the recent proposals of Lambert et al. and recasts them within a classification by DLCQ duality orbits.

The most concrete and checkable by-product is the family of p-brane $T\bar{T}$ flow equations. They explain why the ordinary $T\bar{T}$ flow of 2d Yang–Mills fails to give the D1-brane DBI action. They also give a strikingly simple three-dimensional flow equation, shared by the D2- and M2-brane. These results are useful independently of the string-theory interpretation.

2 Principal drawback

The principal drawback is that the central *holographic* claims are structural and kinematical rather than dynamically verified.

- The DLCQ^n/DLCQ^m correspondence is explicitly a conjecture, supported by pattern-matching across examples (Dp, M2, M5, D1–D5). No quantitative holographic test is offered, such as correlators, spectra or matching of degrees of freedom, and no argument is given for why $m > n$ should be the right condition.
- For the new non-Lorentzian bulk geometries, the authors admit in the Outlook that they have *not* checked the equations of motion and torsional constraints of MpT supergravity. That supergravity is not yet formulated. The geometries are obtained by a "trick" of rescaling ℓ by ω by hand, whose physical meaning (a rescaling of N and of the sources) is left open. Their symmetries and charges are not analysed, and the analysis is restricted to the Poincaré-patch analogue.
- The flow equations are classical, bosonic and abelian. The $p = 3$ equation is given only to linear order in λ, and for $p > 3$ only at zeroth order. The matrix-theory and SYM cases, which are the physical motivation, are therefore not reached.
- The interpretation near asymptotic infinity is itself delicate: non-geometric and quantum effects may dominate there, and for $p > 3$ the theory goes to strong coupling.

In short, the paper is rich in suggestive structure, but its status as a derivation of holography, rather than a systematic classification of limits, rests on unproven assumptions.

3 Most promising follow-up idea

Proposal: extend the $p = 1, 2$ p-brane $T\bar{T}$ flows to the non-abelian (and then supersymmetric) setting, and test them order by order against known non-abelian D-brane effective actions.

What to do. Take the BPS-limit prescription, which defines the flow through the dependence of the brane action on $\lambda = \omega^{-2}$. Apply it to the non-abelian Dp-brane action in a perturbative α' (equivalently λ) expansion, for $p = 1, 2$. The first target is the known F^4 term, which is fixed by the symmetrised-trace structure and by string amplitudes. The second target is the first order at which the symmetrised trace is known to fail (F^6 and commutator-related corrections). The aim is to find out whether a flow equation expressed through the (non-abelian) energy–momentum tensor reproduces these terms, and to identify the obstruction if it does not. For $p = 2$, one would additionally ask whether the same flow acts sensibly on BLG/ABJM, as the authors suggest.

Why this balances feasibility and impact.

- **Feasibility.** The problem is well posed and algebraic, and it has sharp independent checks (the known F^4 result and the known failure of the symmetrised trace at higher order). It can be done perturbatively and does not need an unavailable framework such as MpT supergravity. It builds directly on the existing literature on $T\bar{T}$-like flows to Born–Infeld (Ferko et al.). The authors themselves identify it as an open direction.
- **Impact.** A successful non-abelian extension would turn the paper's most concrete result into a constructive tool. Such a tool would give a symmetry- and flow-based handle on the non-

abelian DBI action, a long-standing open problem. It would also directly realise the paper's claim that undoing the matrix-theory limit deforms SYM to DBI. A negative result (a clear obstruction) would be equally informative about how far the BPS-limit logic extends.

Why not the alternatives. Formulating MpT supergravity and verifying the non-Lorentzian near-horizon solutions would be higher impact for holography, since it would address the main drawback above. However, it is a much larger project, requiring a beta-function or symmetry-based construction. Computing holographic observables to test the DLCQ^n/DLCQ^m conjecture is similarly demanding, because the non-Lorentzian duals lack established dictionaries. These are better treated as longer-term goals, informed by results from the non-abelian flow programme.